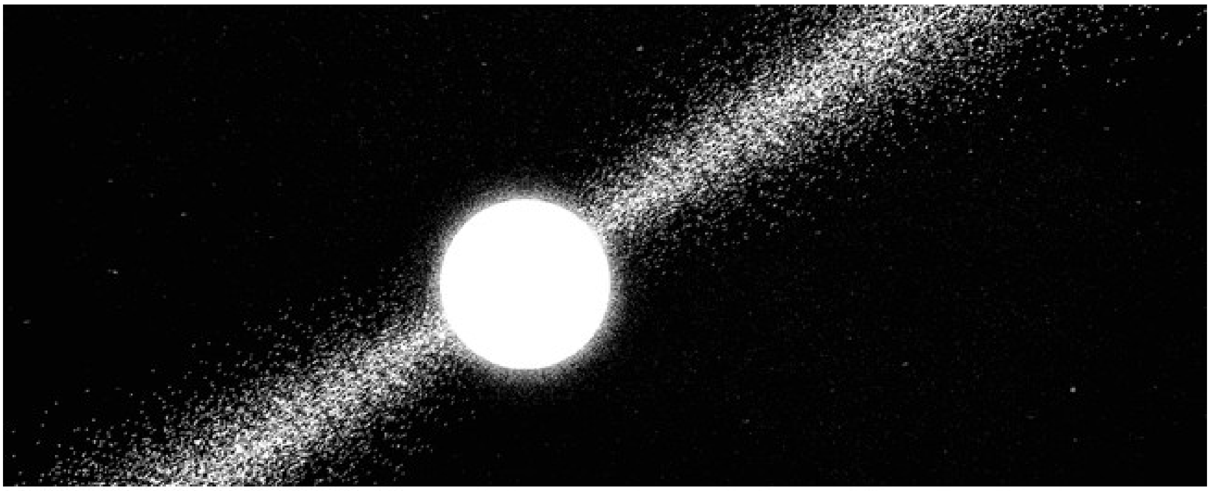
Neutron Stars;
When gravity outweighs atomic Physics



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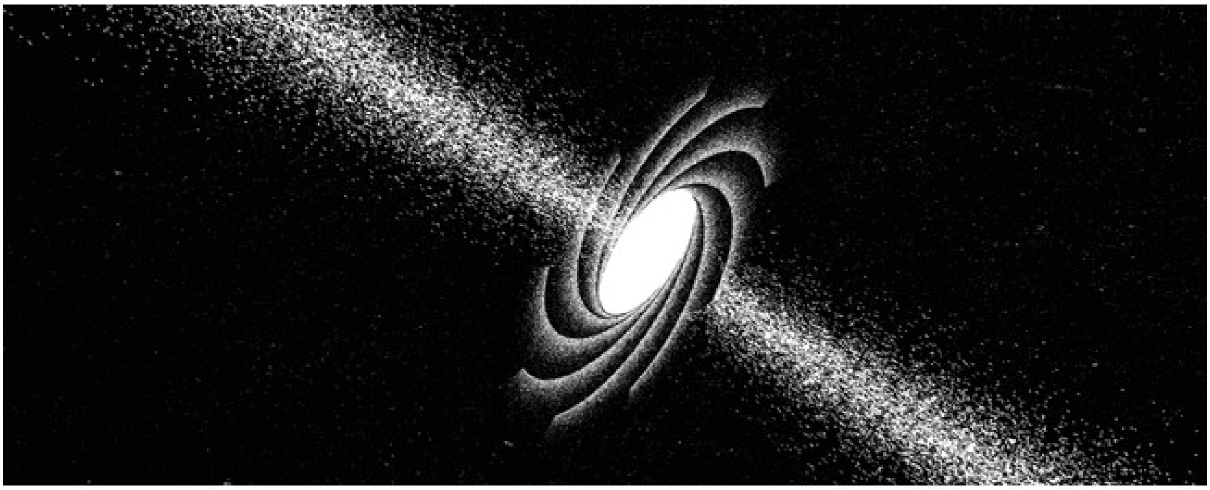

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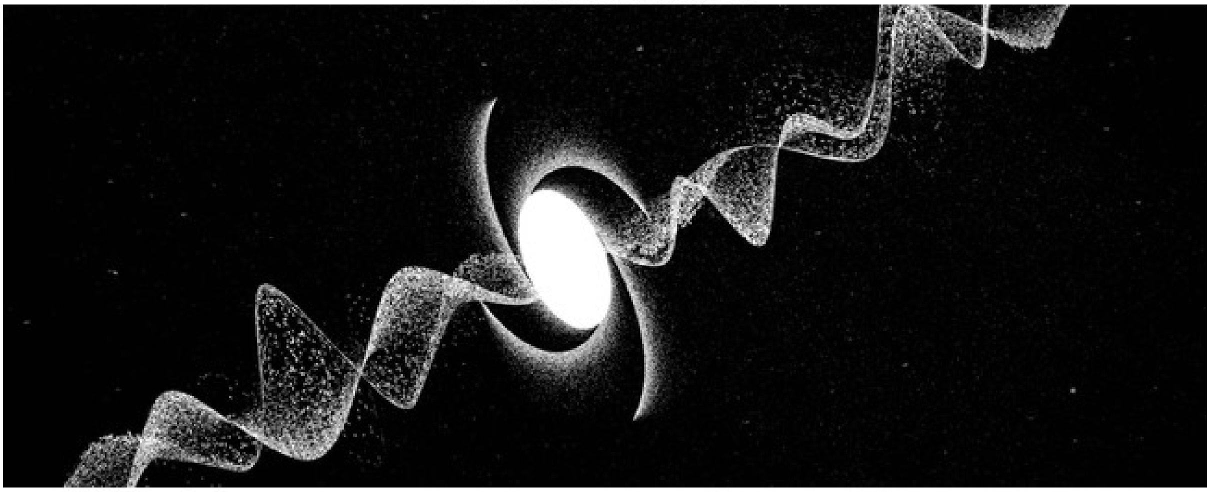

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Introduction & Origin

In this document I want to cover the essentials about neutron stars and why they are fascinating. This section focuses primarily on the fundamental mechanical principles. Although relevant equations are introduced, complex mathematical derivations are omitted to maintain focus on the physical concepts. A Neutron Star is just the super dense core left behind by a massive star after it runs out of fuel and collapse into itself. Not every Star can become one; Our sun (simplified) will just go out (lose its fuel and lose its mass) and leave behind a white dwarf (After it went through the red giant stage), it would have to be around 8-10 $M_{\odot}$heavier to become a Neutron star (or have the chance to), The mass of the collapsing core itself only needs to exceed the Chandrasekhar limit ($\approx 1.4\, M_{\odot}$). When a big Star comes to the end of its life it goes out with a supernova. Massive stars start to burn through their nuclear fuel. The star then starts to turn the core into iron, since iron can't create outward energy by fusion it loses its thermal outward pressure, this results in the entire star collapsing back in on itself and it becomes so dense that Protons and electrons start to merge; They get forced to form neutrons. The outer layers then bounce off and leave the core alone during the Supernova Type II. This is also how a black hole can be made. A neutron star is around 20km in diameter. As the name suggests its almost entirely made of densely packed neutrons. white dwarfs have a theoretical max mass of $\approx 1{,}4\, M_{\odot}$ until they do a type Ia Supernova. This is due to the white dwarf being so dense that electrons go out of their way and therefor produce electron degeneracy pressure which in return this force presses against the gravitation and stabilizes the white dwarf, because it's so dense they move with near light speed. The process that forms the neutrons is called electron capture. The symbol p represents a proton, e^- is an electron, n is a neutron, and ν_e represents an electron neutrino. Inside the collapsing core of a star, extreme gravitational pressure forces electrons into protons to form neutrons and release neutrinos:

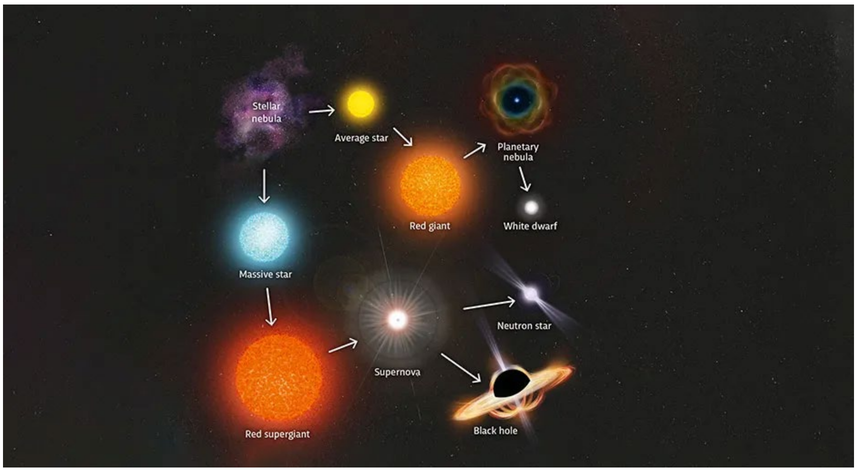

Figure 1; The live cycle of different stars

$p + e^- \longrightarrow n + \nu_e$ This process, as stated before, is known as electron capture (or neutronization) and is what transforms the stellar core into a neutron star. The escaping neutrinos carry away vast amounts of energy, helping drive the supernova explosion. During the collapse the force also accelerates the neutrons stars rotation.

Mass, Size, and Density

A neutron star is right at the limit of collapsing into a black hole (One teaspoon weighs 1 billion to 6 billion tons). To escape its surface, you would need to travel at around $0.4c$ to $0.5c$. People often assume bigger always means heavier and scale does matter for gravity; if two objects share the same uniform density, a larger object will have a stronger surface gravity than a smaller one because surface gravity scales with both density and size however neutron stars actually become smaller and denser as they gain mass, meaning surface gravity scales much more sharply due to stronger compactification.

Tolman-Oppenheimer-Volkoff (TOV)

The Tolman-Oppenheimer-Volkoff (TOV) equation describes the maximum mass a neutron star can have before it collapses into a black hole. It is calculated using the following differential equation:

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}\left(1 + \frac{P(r)}{\rho(r)c^2}\right)\left(1 + \frac{4\pi r^3 P(r)}{M(r)c^2}\right)\left(1 - \frac{2GM(r)}{c^2 r}\right)^{-1}$$

This equation shows how much pressure needs to decrease towards the outside ($\frac{dP}{dr}$) to fight against the insane gravitation, pulling everything towards the center.

Breakdown of the Components

- $-\frac{GM(r)\rho(r)}{r^2}$: This part is just the original, well-known Newtonian formula for hydrostatic equilibrium, where G is the gravitational constant, $M(r)$ is the enclosed mass at radius r, and $\rho(r)$ is the mass density at radius r.
- $\left(1 + \frac{P(r)}{\rho(r)c^2}\right)$: In General Relativity (GR), not only mass attracts and pulls other objects, but also energy and pressure. High pressure (P) therefore strengthens the gravitational pull relative to the rest-mass energy density (ρc^2).
- $\left(1 + \frac{4\pi r^3 P(r)}{M(r)c^2}\right)$: This term represents the equivalent mass contribution from the total pressure contained within the spherical volume of radius r.
- $\left(1 - \frac{2GM(r)}{c^2 r}\right)^{-1}$: This is taken from the Schwarzschild metric and defines the geometry of space and time inside the star, accounting for radial space curvature caused by mass-energy concentration.

Step-by-Step Calculation Process

To model a complete star, the TOV equation is not solved in a single algebraic calculation, but as a coupled system of differential equations integrated numerically from $r = 0$ (the center) out to $r = R$ (the surface); (See next page)

1. Setting the Initial Boundary Conditions ($r = 0$):
    - Choose a central density ρ_0 and corresponding central pressure P_0.
    - Set the enclosed mass at the centre to zero: $M(0) = 0$.
2. Linking State Variables via Equation of State (EoS):
    - The TOV equation has three unknown functions: $P(r)$, $\rho(r)$, and $M(r)$.
    - To solve it, physicists supply an Equation of State $P = P(\rho)$, which relates local density to local pressure based on nuclear physics models.
3. Updating the Enclosed Mass Step-by-Step:
    - As you move outward by a small distance step dr, the mass grows according to the mass-continuity equation in GR:

$$\frac{dM}{dr} = 4\pi r^2 \rho(r)$$

    - At each distance r, the new mass $M(r)$ is calculated by integrating the density of the shell at that radius.
4. Calculating the Pressure Gradient ($\frac{dP}{dr}$):
    - Plug r, $M(r)$, $P(r)$, and $\rho(r)$ into the TOV equation to find the slope $\frac{dP}{dr}$.
    - The negative sign guarantees that pressure decreases as distance from the core increases.
5. Stepping Outward to the Surface ($P = 0$):
    - Numerical integration (e.g., using a Runge-Kutta algorithm) computes the new pressure $P(r + dr) = P(r) + \frac{dP}{dr} dr$.
    - The integration continues outward step-by-step until the pressure drops to zero ($P = 0$). The radius where $P = 0$ marks the total radius (R) of the star, and the mass enclosed at that radius $M(R)$ is the total mass (M).

By repeating this step-by-step integration for higher central densities (ρ_0), the total calculated mass reaches a maximum peak. That peak is the TOV mass limit.

This equation shows that the maximum mass for a neutron star is typically between 2.0 and 2.3 solar masses ($M_\odot$). Most neutron stars have a mass of approximately 1.4 $M_\odot$, and the density in their core reaches around 10^{17} kg/m^3 (exceeding nuclear saturation density).

Quantum Mechanics & Degeneracy Pressure

The reason a neutron star does not collapse into a black hole is neutron degeneracy pressure, which is driven by the Pauli Exclusion Principle. Two Neutrons are not allowed to experience the same quantum state at the same time. We will look at this equation;

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

Because its spread across such a small space (Δx), the impulse (Δp) gradually increases. $\hbar$ represents the reduced Planck constant, which scales quantum effects and equals Planck's constant ($\hbar$) divided by 2π ($\hbar = \frac{h}{2\pi}$). This results in an extreme outward pressure. At extremely short distances ($< 0{,}7$ fm or 0.7×10^{-15} meters), the strong nuclear force becomes repulsive, additionally helping to prevent the collapse. Also, because the Neutron star is rotating so quickly it additionally helps to stretch it outward however it is neutron degeneracy pressure and the repulsive core strong force that actively halt gravitational collapse, not rotation.

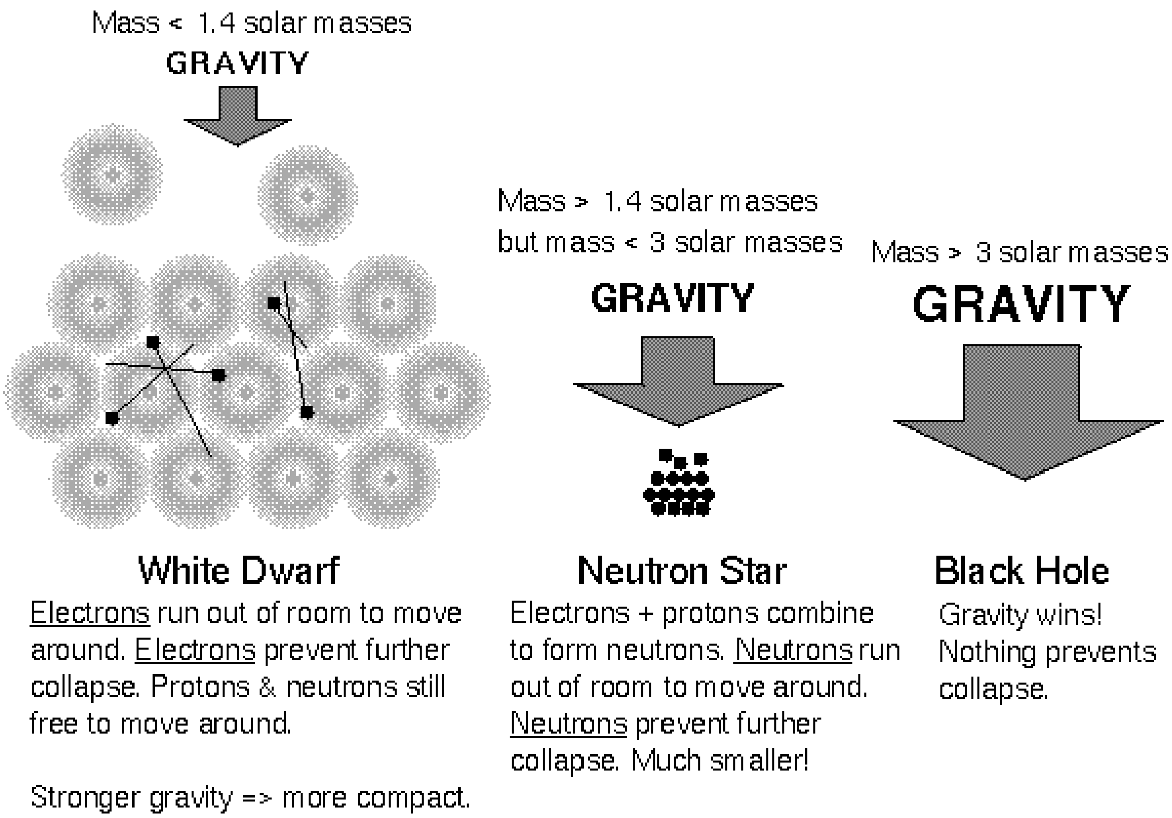

Figure 2; Collapses on different scale levels

As you can see on Figure 2; if the neutron star exceeds the TOV limit it collapses, our sun doesn't collapse and turns into a white dwarf.

Internal Structure & "Nuclear Pasta”

The outer crust of a neutron star is a couple hundred meters thick; It consists of a rigid crystal lattice of heavy atomic nuclei (such as iron) immersed in a dense gas of relativistic electrons. The inner crust reaches deeper toward the core, where densities surpass the neutron drip point. Here, nuclei become heavily neutron-rich, and free neutrons leak out to form a quantum liquid superfluid alongside a degenerate electron gas.

As depth and density increase toward the bottom of the inner crust, the competing forces of short-range nuclear attraction and long-range proton repulsion cause atomic nuclei to lose their spherical shapes, melting and merging into complex structures:

Gnocchi (Droplets): Isolated, pseudo-spherical clusters of nucleons at the upper boundary of the inner crust.

Spaghetti (Rods): Long, parallel cylinders of nuclear matter formed when droplets merge under higher compression.

Lasagna (Slabs/Plates): Flat, infinite sheets of nuclear matter stacked on top of each other. Computer models suggest nuclear lasagna may be up to 10 billion times stronger than steel.

Anti-spaghetti (Tubes): Cylindrical holes or tubes of neutron-fluid running through a continuous block of nuclear matter (the inverse of the spaghetti phase).

Swiss Cheese (Bubbles): Spherical voids or holes punched through dense nuclear planes.

Waffles: Complex, grid-like periodic depressions and saddle surfaces discovered via advanced molecular-dynamics simulations.

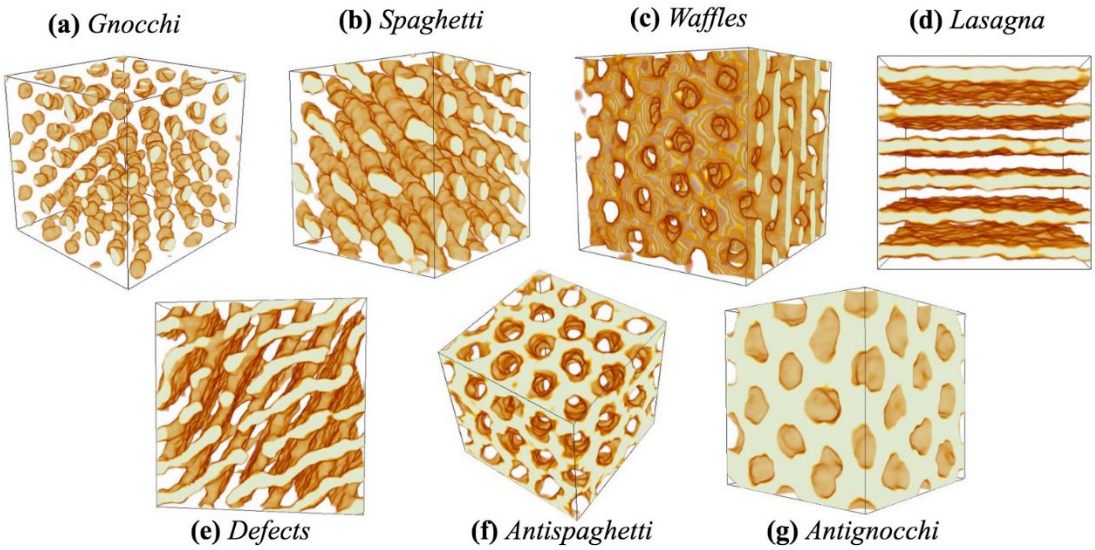

Figure 3; Pasta inside of a neutron star

The outer core begins where the nuclear pasta completely dissolves and uniform nuclear matter takes over. It is typically several kilometres thick.

Composition: It is primarily a dense soup of free neutrons, with a small percentage (around 5%) of protons and electrons.

Superfluidity: The neutrons form a Cooper-paired quantum superfluid, meaning they flow with absolutely zero friction or viscosity.

Superconductivity: The protons pair up to form a type-II superconductor, creating a highly conductive environment that locks in the star's massive magnetic fields.

The inner core occupies the very centre of the star, spanning the final few kilometres where densities exceed several times the density of an atomic nucleus. The exact state of matter here remains one of the biggest mysteries in modern astrophysics. Scientists have proposed several competing models:

Hyperon Core: At extreme pressures, ordinary neutrons and protons may transform into exotic, heavy particles containing strange quarks, known as hyperons.

Pion or Kaon Condensate: The high density might cause mesons (like pions or kaons) to form a dense, uniform Bose-Einstein condensate at the centre.

Deconfined Quark Matter: The boundaries of individual neutrons and protons may completely break down, melting them into a free-flowing "soup" of up, down, and strange quarks (a quark-gluon plasma).

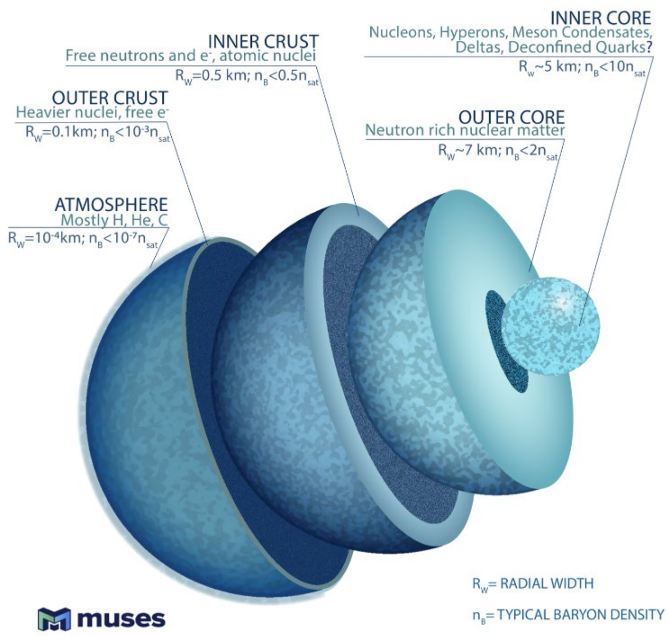

Figure 4; The inside of a neutron star

Rotation, Magnetism & Pulsars

As mentioned before; neutron stars rotate around their own axis with a fast speed of 1×10^1 to 1×10^2 rotations per second. Some can spin even faster; the fastest known pulsar spins at 716 Hz (716 rotations per second). The reason why they rotate so fast is the conservation of angular momentum. The total angular momentum (L) of a spinning object must remain constant if no external torque acts on it.

$$L = I\omega = \text{constant}$$

Here, I represents the moment of inertia and ω represents the angular velocity. For a spherical star, the moment of inertia is proportional to the square of its radius ($I \propto R^2$).

During the collapse of a massive star, its radius shrinks drastically. For instance; from 10^6 km to only 10 km. The radius R decreases by a factor of $100{,}000$, which causes the moment of inertia I to drop by a factor of 10^{10}. To keep the angular momentum L constant, the angular velocity ω must increase dramatically, forcing the neutron star to spin hundreds of times per second.

The extreme magnetic field can be explained using the magnetic flux conservation.

$$\Phi = \int B \cdot dA \approx B \cdot R^2 = \text{constant}$$

As the star contracts, its surface area decreases drastically. Because $B \propto 1/R^2$, compressing the star's original magnetic field into a sphere only 10 km wide amplifies the magnetic field strength tremendously. This process yields magnetic fields ranging from 10^4 to 10^8 Tesla. In extreme cases, known as magnetars even 10^9 to 10^{11} Tesla.

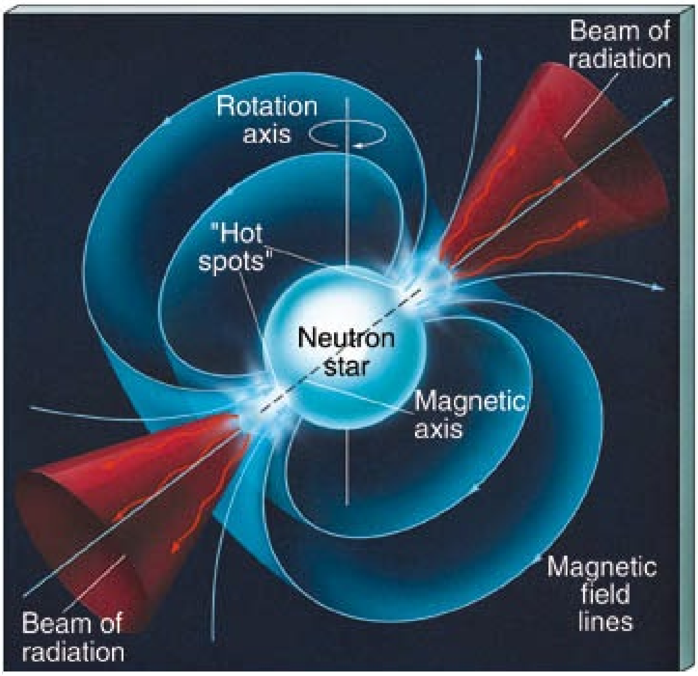

Figure 5; Rotation of a neutron star

Collisions & The Origin of Heavy Elements (Kilonovae)

Since Neutron stars have such a strong gravitational pull, they could pull in and collide with another neutron star. If they orbit each other, they both radiate energy outward (gravitational waves) which is ripples in the fabric of space time. As energy drains from, they're system, the orbit steadily decays, causing them both to spiral inward, until they collide at a fraction of the speed of light. This triggers an intense explosion known as a Kilonova which is around 1000 times bigger than a standalone nova. The energy loss which causes the energy merger is calculated with GR. In 2017, the LIGO and Virgo observatories detected event GW170817, capturing gravitational waves from a neutron star merger alongside electromagnetic counterparts (GRB 170817A), directly confirming kilonova models. During the violent impact, massive amounts of neutron-dense fluid are ejected into space at high speeds ($0.1c$–$0.3c$). Because the free-neutron density exceeds 10^{20} neutrons/cm^3, seed atomic nuclei capture free neutrons far faster than they can radioactively decay ($\tau_{\text{capture}} \ll \tau_{\beta\text{-decay}}$). This rapid accumulation builds highly unstable, ultra-heavy nuclei well beyond the valley of stability. Over subsequent days and weeks, these exotic isotopes undergo chains of beta minus (β^-) decay:

$$\mathrm{n} \longrightarrow \mathrm{p} + \mathrm{e}^- + \bar{\nu}_e$$

As neutrons inside the newly formed nuclei convert into protons, the atomic number increases, synthesizing stable heavy elements including Gold ($Z = 79$), Platinum ($Z = 78$), and Uranium ($Z = 92$). The radioactive energy released by this ongoing decay illuminates the surrounding debris, producing the characteristic thermal glow of a kilonova.

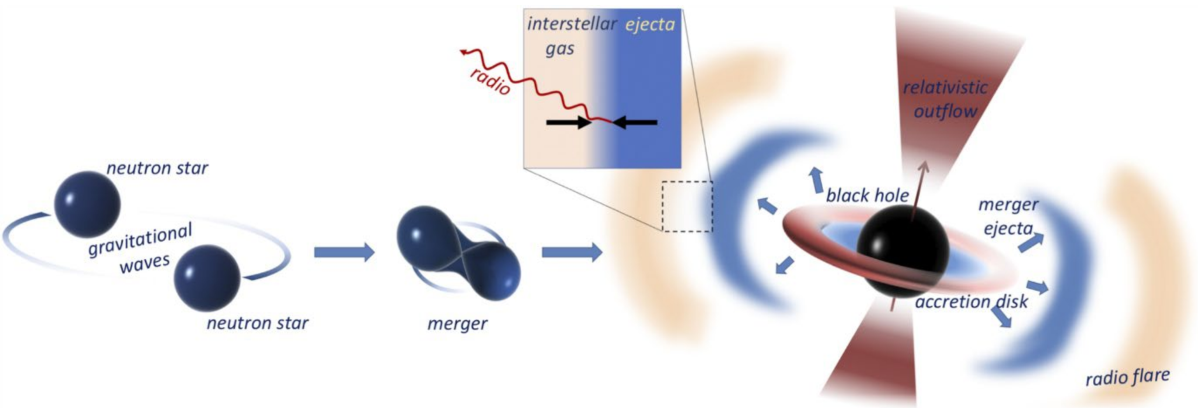

Figure 6; Neutron stars merging

Conclusion and Einstein@Home

Neutron stars serve as ideal cosmic laboratories for testing General Relativity and quantum mechanics, as they allow us to observe how immense amounts of mass behave in an extremely confined space. They represent the physical limit of stable mass in the universe, approaching the maximum Tolman-Oppenheimer-Volkoff (TOV) limit before a collapse into a black hole becomes inevitable. Understanding neutron stars could unlock key answers to how matter behaves under extreme conditions and deepens our understanding of black holes.

Einstein@Home

To locate and study these extreme objects, initiatives like **Einstein@Home** rely on public participation. It is a volunteer computing project that uses idle processing power from participants' personal computers to analyse astronomical data in search of neutron stars. Over the past 20 years, the project has successfully discovered around 90 new neutron stars. Participating contributes directly to real scientific progress and in astrophysics, confirming where an object *is not* just as valuable as finding where it is.

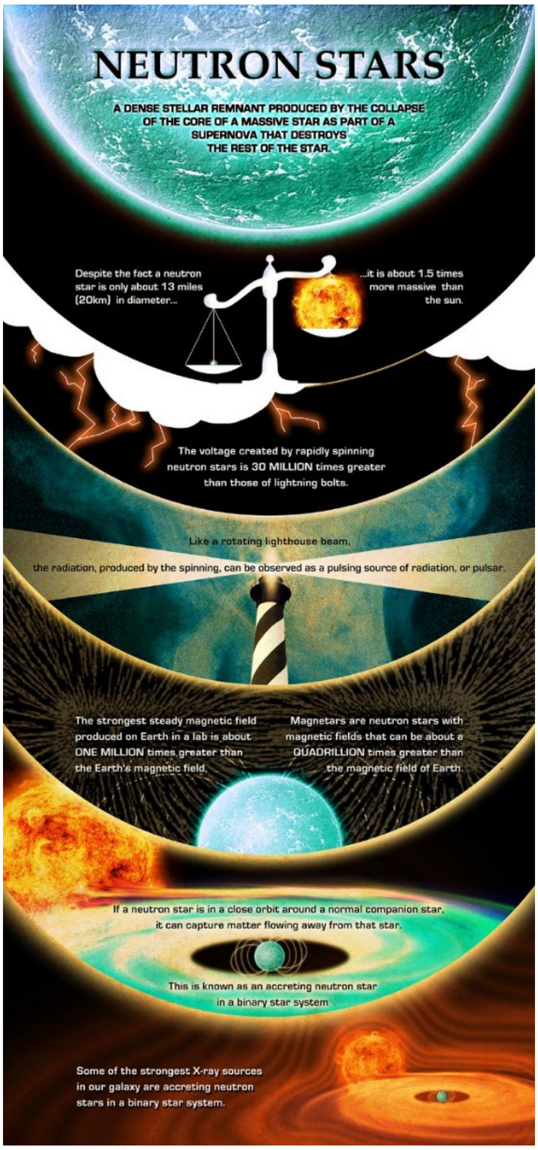

Figure 7; Interesting facts about neutron stars

In just 14 to 16 hours of running the software, my own computer completed roughly 1.01 quintillion (1.01×10^{18}) floating-point operations for the project. Since numbers can be hard to comprehend, how much is that actually? If every person out of 8 billion people solved one complex math problem every second without stopping, the entire global population would need about 4 years (Its exactly 4.00 years of non-stop work.) of continuous collective effort to finish 1.01 quintillion operations. If you were to count from 1 to 1.01 quintillion at a rate of one number per second, it would take you roughly 32 billion years, which is more than twice $(\approx 2.32\,\times)$ the age of our universe. These numbers highlight how fundamentally different human intuition is from cosmic reality. What seems like an incomprehensibly vast timeframe to us is practically instantaneous on a universal scale. Projects like Einstein@Home bridge this gap, allowing public collaboration to drive real astronomical discovery.